

1

$1 \leq T \leq 10$
 $1 \leq N \leq 10^4$
 $1 \leq u, v \leq N$

2

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input

1
1



Sample output

25



Explanation

The first line represents the number of test cases, $T = 1$

For test case 1:

Since there is only 1 node, You can assign any prime number to it

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F4



F5

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button).

1

2

- The first line contains a single integer T , which denotes the number of test cases. T also specifies the number of times you have to run the `validAssignment` function on a different set of inputs.
- For each test case:
 - The first line contains an integer N .
 - The next $N - 1$ line contains 2 space-separated integers denoting an edge.

Output format

For each test case, print the total number of valid assignments modulo $10^9 + 7$ in a new line.

Constraints

$$1 \leq T \leq 10$$

$$1 \leq N \leq 10^4$$

$$1 \leq u, v \leq N$$

Code snippets (also called starter code/boilerplate code)

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Prime tree

2

You are given an undirected tree T consisting of N nodes rooted at node 1. You have to assign value to each node of the tree such that:

- The assigned value must be a prime number less than or equal to 10^2 .
- The Sum of values assigned to any pair of adjacent nodes is not a prime number.

Task

Determine the total number of valid assignments. Since, this number can be very large output it modulo $10^9 + 7$.

Example

Assumptions

- $T = 1$
- $N = 2$
- $edges = [(1,2)]$

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Explanation

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You have to deliver 3 items and the minimum delivery price among the three items is 4, hence the delivery price charged for the rest two items is also 4.

Total price = $(6 + 2 + 1) + 4 \times 2 = 21$

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Time Limit: 2.0 sec(s) for each input file
Memory Limit: 256 MB
Source Limit: 1024 KB

Scoring

Score is assigned if any testcase passes

Allowed Languages

Bash, C, C++14, C++17, Clojure, C#, D, Erlang, F#, Go, Groovy, Haskell, Java 8, JavaScript(Node.js), Julia, Kotlin, Lisp (SBCL), Lua, Objective-C, OCaml, Octave, Pascal, Perl, PHP, Python, Python 3, R(RScript), Racket, Ruby, Rust, Scala, Swift, TypeScript, Visual Basic

C (gcc 10.3)

Auto-complete ready!

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int solve (int n, int m, int** arr) {
6     // Write your code here
7
8 }
9
10 int main() {
11     int n;
12     scanf("%d", &n);
13     int m;
14     scanf("%d", &m);
15     int i_arr, j_arr;
16     int **arr = (int **)malloc((n)*sizeof(int *));
17     for(i_arr = 0; i_arr < n; i_arr++)
18     {
19         arr[i_arr] = (int *)malloc((2)*sizeof(int));
20     }
21     for(i_arr = 0; i_arr < n; i_arr++)
22     {
23         for(j_arr = 0; j_arr < 2; j_arr++)
24         {
25             scanf("%d", &arr[i_arr][j_arr]);
26         }
27     }
28 }
```

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View test cases ^

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Input format

- The first line contains two space-separated integers n and m respectively.
- The next n lines contain two space-separated integers c_i and d_i denoting the price and the delivery price of the item respectively.

Output format

Print the maximum amount of money that you get after delivering m items. The output must be printed in a single line.

Constraints

$$1 \leq m \leq n \leq 10^5$$
$$1 \leq c_i, d_i \leq 10^6$$

Sample input	Sample output
3 3 6 8 2 5 1 4	21

C (gcc 10.3)

Auto-complete ready!

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int solve (int n, int m, int** arr) {
6     // Write your code here
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10 int main() {
11     int n;
12     scanf("%d", &n);
13     int m;
14     scanf("%d", &m);
15     int i_arr, j_arr;
16     int **arr = (int **)malloc((n)*sizeof(int *));
17     for(i_arr = 0; i_arr < n; i_arr++)
18     {
19         arr[i_arr] = (int *)malloc((2)*sizeof(int));
20     }
21     for(i_arr = 0; i_arr < n; i_arr++)
22     {
23         for(j_arr = 0; j_arr < 2; j_arr++)
24         {
25             scanf("%d", &arr[i_arr][j_arr]);
26         }
27     }
28 }
```

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Question 2

Max. score: 30.00

Maximizing profit

You have decided to start an e-commerce business because there is an opportunity to grow the business in this crisis of products due to lockdown.

You have n items that must be delivered to a customer. Here, the price of the i^{th} item is denoted by c_i and the delivery price of that item is denoted by d_i . If a customer orders more than one item, then the delivery price of each item is the minimum of the delivery price of all items ordered.

You can only deliver exactly m items out of all the items. You have a limited amount of money, therefore, you want to select m items in a way that maximizes the total sum of money that you get after the successful delivery of items. Refer to the explanation for more details.

Input format

- The first line contains two space-separated integers n and m respectively.
- The next n lines contain two space-separated integers c_i and

C (gcc 10.3)

Auto-complete ready!

```

1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int solve (int n, int m, int** arr) {
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15     int i_arr, j_arr;
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17     for(i_arr = 0; i_arr < n; i_arr++)
18     {
19         arr[i_arr] = (int *)malloc((2)*sizeof(int));
20     }
21     for(i_arr = 0; i_arr < n; i_arr++)
22     {
23         for(j_arr = 0; j_arr < 2; j_arr++)
24         {
25             scanf("%d", &arr[i_arr][j_arr]);
26         }
27     }
28 }

```

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